

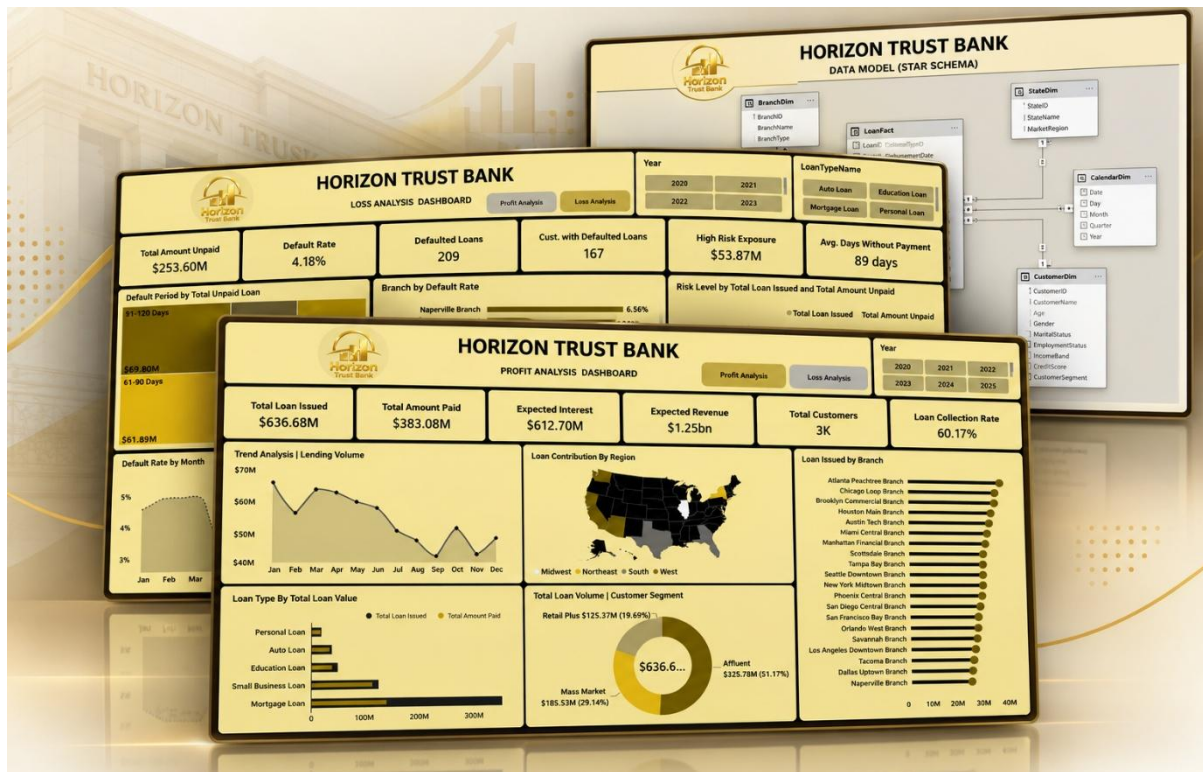
Horizon Trust Bank: Loan Portfolio Performance & Credit Risk Analysis Using Power BI

How I used data modelling, DAX and Power BI to analyse a \$636M loan portfolio, uncover repayment patterns, and identify areas of credit-risk exposure.

by

Data Analyst: Oluwatosin Hamzat

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Introduction

A bank can have a large and growing loan portfolio while still carrying significant financial risk.

For a lending institution, knowing how much money has been issued is only part of the story. Management also needs to understand how much has been collected, how much remains unpaid, where defaults are concentrated, which loan products carry the greatest exposure, and which branches require closer attention.

That was the business problem behind this project for **Horizon Trust Bank**, a mid-sized commercial bank operating across multiple U.S. states through urban, retail, and commercial branches. The bank offers a diversified lending portfolio covering personal, mortgage, auto, small business, and education loans.

The objective was to transform the bank's loan data into an integrated Power BI solution that would allow management to monitor **loan performance, repayment behaviour, profitability indicators, defaults, and credit-risk concentration.**

The project brief specifically called for a structured data model and interactive dashboard capable of connecting customer, loan, branch, product, geographical and risk information.

The Business Problem

Horizon Trust Bank had a growing loan portfolio, but the underlying data was not structured for effective analysis.

This made it difficult to understand loan profitability, identify potential losses, monitor defaults, and determine where credit risk was concentrated.

Management needed a consolidated view that could answer questions such as:

- How much has the bank lent?
- How much has actually been collected?
- Which loan products account for the largest lending exposure?
- Where is unpaid loan exposure concentrated?
- Which branches have the highest default rates?
- How has lending volume changed over time?
- Which customer segments carry the greatest exposure?
- How much of the portfolio is classified as high risk?
- How long are loans remaining unpaid?

The project brief highlighted the need to identify credit-risk concentration, compare branch and state performance, evaluate customer risk profiles, assess collateral adequacy, and understand repayment and default patterns.

So, rather than simply creating a dashboard, I approached the project as a loan portfolio performance and credit-risk analysis problem.

Project Objectives

I structured the project around four major objectives:

1. Build an integrated loan portfolio model

Connect loan transactions with customer, branch, geography, loan type, collateral and calendar information.

2. Measure lending performance

Track loan value, repayments, expected interest, expected revenue and collection rate.

3. Analyse credit risk

Identify default rates, unpaid exposure, high-risk loans, repayment delays and branch-level risk.

4. Turn analysis into an interactive management dashboard

Allow management to filter the portfolio by year, loan type and other dimensions while monitoring both performance and risk.

These objectives align with the project initiative to build a robust star schema, visualize credit-risk concentration, compare branch/state performance, and analyse repayment behaviour and collateral coverage.

Data Preparation

Before building the dashboard, the data needed to be structured for analysis.

The workflow followed five stages:

- **Clean and standardize the raw loan data**
- **Construct the star-schema data model**
- **Create KPIs and DAX measures**
- **Design the Power BI dashboard**
- **Interpret the results and translate them into business insights**

This workflow follows the project specification provided for the case study. I also created a calendar table covering the analysis period, allowing the loan portfolio to be analysed across years and months.

Data & Data Modelling

The dataset contains **5,000 loan records** and **2,773 unique customers**.

The central fact table is **LoanFact**, supported by several dimension tables:

- Customer Dimension
- Branch Dimension
- State Dimension
- Loan Type Dimension
- Collateral Type Dimension
- Calendar Dimension

LoanID	LoanAccountNumber	CustomerID	BranchID	StateID	LoanTypeID	CollateralTypeID	DisbursementDate	MaturityDate	LastPaymentDate	LoanTermMonths	InterestRate	OriginalLoanAmount	ScheduleMonthlyInstallment	TotalAmountPaid	CollateralValue	PerformanceScore	MissedPaymentFlag	DefaultFlag	CollectionOutcome	RiskLevel	LoanStatus	LoanDefaultPeriod
LN-000129	L-7200131	CU-000129	BR-017	ST-45	LT-01	CT-01	14 November 2022	14 November 2027	28 June 2026	48	9.93	30217.3	763.22	23474.61	0	43.96	Yes	No	Partial Payment	Medium	Active	61-90 Days
LN-000130	L-7200132	CU-000130	BR-006	ST-43	LT-01	CT-01	10 August 2022	10 August 2025	20 June 2026	36	12.23	16480.86	546.78	16165.56	0	40.88	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-000135	L-7200135	CU-000145	BR-005	ST-43	LT-01	CT-01	22 June 2020	22 June 2022	15 July 2026	24	12.21	16712.6	769.49	15824.19	0	37.2	Yes	No	Partial Payment	Medium	Active	61-90 Days
LN-000187	L-7200187	CU-000191	BR-001	ST-41	LT-01	CT-01	07 April 2023	07 April 2028	31 May 2026	60	11.31	19824.83	434.33	17566.47	0	31.42	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-000192	L-7200192	CU-000412	BR-015	ST-43	LT-01	CT-01	12 February 2021	12 February 2024	26 July 2026	36	10.42	28952.82	840.18	28126.64	0	57.6	Yes	No	Partial Payment	Medium	Active	31-60 Days
LN-000178	L-7200178	CU-000192	BR-013	ST-41	LT-01	CT-01	28 March 2024	28 March 2027	07 May 2026	36	11.4	19943.8	637.34	17886.34	0	32.14	Yes	No	Partial Payment	Medium	Active	121-150 Days
LN-000178	L-7200178	CU-000195	BR-012	ST-48	LT-01	CT-01	03 June 2022	03 June 2027	27 May 2026	60	12.25	17586.29	393.42	16421.63	0	36.48	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-000179	L-7200179	CU-000165	BR-017	ST-45	LT-01	CT-01	29 June 2023	29 June 2028	09 June 2026	60	13.62	13384.72	313.42	11914.45	0	33.62	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-000200	L-7200200	CU-000124	BR-020	ST-48	LT-01	CT-01	22 May 2020	22 May 2022	03 August 2026	24	11.14	31706.51	1479.83	31188.1	0	46.48	Yes	No	Partial Payment	Medium	Active	31-60 Days
LN-000226	L-7200226	CU-000135	BR-003	ST-42	LT-01	CT-01	25 January 2020	25 January 2025	11 May 2026	60	9.59	30946.56	646.03	29126.66	0	52.26	Yes	No	Partial Payment	Medium	Active	121-150 Days
LN-000274	L-7200274	CU-000161	BR-013	ST-41	LT-01	CT-01	25 April 2023	25 April 2025	18 June 2026	24	12.45	17224.53	813.5	16266.51	0	35.62	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-000338	L-7200338	CU-000167	BR-020	ST-48	LT-01	CT-01	12 June 2022	12 June 2026	23 August 2026	48	10.07	30365.52	786.41	29027.87	0	57.04	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-000339	L-7200339	CU-000119	BR-007	ST-04	LT-01	CT-01	02 January 2020	02 January 2025	14 July 2026	60	13.45	12817.08	294.61	12145.47	0	34.66	Yes	No	Partial Payment	Medium	Active	61-90 Days
LN-000400	L-7200400	CU-000113	BR-013	ST-41	LT-01	CT-01	31 January 2021	31 January 2025	06 June 2026	48	9.99	29202.01	741.01	28636.35	0	50.46	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-000438	L-7200438	CU-000175	BR-015	ST-43	LT-01	CT-01	22 March 2023	22 March 2025	01 July 2026	24	11.06	22944.64	1116.54	22325.23	0	22.58	Yes	No	Partial Payment	Medium	Active	61-90 Days
LN-000451	L-7200451	CU-000199	BR-013	ST-41	LT-01	CT-01	05 July 2023	05 July 2025	27 July 2026	24	12.63	18334.04	876.5	17391.54	0	26.04	Yes	No	Partial Payment	Medium	Active	31-60 Days
LN-000481	L-7200481	CU-000162	BR-009	ST-45	LT-01	CT-01	14 September 2022	14 September 2023	14 August 2026	12	12.24	15486.51	1200.85	12047.41	0	37.36	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-000546	L-7200546	CU-000162	BR-009	ST-45	LT-01	CT-01	16 August 2022	16 August 2025	03 May 2026	36	10.65	30176.35	891.65	28588.21	0	45.42	Yes	No	Partial Payment	Medium	Active	121-150 Days
LN-000554	L-7200554	CU-000169	BR-018	ST-46	LT-01	CT-01	25 November 2020	25 November 2023	12 August 2026	36	11.08	32883.12	1010.67	29263.66	0	45.72	Yes	No	Partial Payment	Medium	Active	31-60 Days
LN-000580	L-7200580	CU-000176	BR-006	ST-43	LT-01	CT-01	14 October 2021	14 October 2025	06 August 2026	48	11.35	19270.42	500.01	18488.3	0	26.52	Yes	No	Partial Payment	Medium	Active	31-60 Days
LN-000631	L-7200631	CU-000165	BR-019	ST-47	LT-01	CT-01	24 January 2021	24 January 2025	02 June 2026	48	11.67	19371.51	507.02	18851.69	0	42.38	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-000670	L-7200670	CU-000187	BR-010	ST-46	LT-01	CT-01	27 December 2022	27 December 2027	22 July 2026	60	11.89	31174.81	679.22	29707.73	0	45.06	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-000709	L-7200709	CU-000260	BR-013	ST-41	LT-01	CT-01	16 January 2021	16 February 2023	22 June 2026	24	9.75	31873.48	1487.12	29385.79	0	57.48	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-000134	L-7200134	CU-000450	BR-040	ST-48	LT-01	CT-01	10 December 2021	10 December 2026	18 May 2026	60	16.43	14403.61	480.3	14403.61	0	40.06	Yes	No	Partial Payment	Medium	Active	121-150 Days
LN-000766	L-7200766	CU-000103	BR-017	ST-45	LT-01	CT-01	07 April 2022	07 April 2025	29 June 2026	36	11.63	18876.76	595.24	18594.07	0	45.24	Yes	No	Partial Payment	Medium	Active	61-90 Days
LN-000823	L-7200823	CU-000263	BR-010	ST-46	LT-01	CT-01	04 May 2021	04 May 2023	06 June 2026	24	12.7	19648.55	931.41	19188.87	0	26.32	Yes	No	Partial Payment	Medium	Active	61-90 Days
LN-000880	L-7200880	CU-0001616	BR-006	ST-43	LT-01	CT-01	07 August 2022	07 August 2027	18 July 2026	60	12.33	16577.67	371.53	15886.57	0	48.24	Yes	No	Partial Payment	Medium	Active	61-90 Days
LN-000901	L-7200901	CU-000548	BR-003	ST-42	LT-01	CT-01	02 January 2024	02 January 2026	15 June 2026	24	10.47	28812.23	1235.94	28532.38	0	49.2	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-000913	L-7200913	CU-000261	BR-019	ST-47	LT-01	CT-01	07 May 2021	07 May 2022	29 July 2026	12	11.03	23802.05	2108.47	20748.29	0	41.2	Yes	No	Partial Payment	Medium	Active	31-60 Days
LN-000914	L-7200914	CU-000091	BR-009	ST-45	LT-01	CT-01	27 June 2023	27 June 2024	06 June 2026	12	9.6	25383.55	2200.58	22188.06	0	50.38	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-000915	L-7200915	CU-000126	BR-018	ST-46	LT-01	CT-01	13 May 2024	13 May 2028	23 May 2026	48	11.1	19982.26	516.93	12776.46	0	44	Yes	No	Partial Payment	Medium	Active	121-150 Days
LN-000932	L-7200932	CU-000262	BR-010	ST-46	LT-01	CT-01	17 May 2022	17 May 2026	27 June 2026	48	10.54	31486.75	804.73	29047.16	0	30.32	Yes	No	Partial Payment	Medium	Active	61-90 Days
LN-000939	L-7200939	CU-000859	BR-010	ST-46	LT-01	CT-01	28 August 2023	28 August 2024	13 June 2026	12	13.69	16848.3	1482.47	15329.43	0	17.58	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-000928	L-7200928	CU-000262	BR-005	ST-43	LT-01	CT-01	24 June 2021	24 June 2022	05 June 2026	12	10.63	24852.6	2203.92	24533.06	0	18.4	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-001030	L-7201030	CU-000388	BR-020	ST-48	LT-01	CT-01	11 May 2021	11 May 2024	15 July 2026	36	10.43	30153.57	894.83	28198.83	0	36.28	Yes	No	Partial Payment	Medium	Active	61-90 Days
LN-001031	L-7201031	CU-000165	BR-014	ST-42	LT-01	CT-01	18 June 2024	18 June 2026	30 June 2026	24	10.67	29907.27	1388.34	29469.69	0	54.3	Yes	No	Partial Payment	Medium	Active	61-90 Days
LN-001052	L-7201052	CU-000117	BR-013	ST-41	LT-01	CT-01	24 March 2024	24 March 2025	22 August 2026	12	13.54	12759.26	1142.86	12467.07	0	35.02	Yes	No	Partial Payment	Medium	Active	31-60 Days
LN-001113	L-7201113	CU-000963	BR-013	ST-41	LT-01	CT-01	01 May 2020	01 May 2025	25 July 2026	60	10.55	22689.05	493.62	22215.96	0	26.78	Yes	No	Partial Payment	Medium	Active	31-60 Days
LN-001131	L-7201131	CU-000170	BR-015	ST-43	LT-01	CT-01	30 December 2020	30 December 2021	07 August 2026	12	11.28	18449.09	1724.32	18886.1	0	24.2	Yes	No	Partial Payment	Medium	Active	31-60 Days
LN-001137	L-7201137	CU-000485	BR-013	ST-41	LT-01	CT-01	20 January 2024	20 January 2025	11 May 2026	12	11.06	26325.41	2586.2	27894.74	0	27.04	Yes	No	Partial Payment	Medium	Active	121-150 Days
LN-001176	L-7201176	CU-000179	BR-004	ST-42	LT-01	CT-01	14 July 2022	14 July 2025	27 June 2026	36	10.54	30246.64	983.66	28762.03	0	22.88	Yes	No	Partial Payment	Medium	Active	61-90 Days
LN-001204	L-7201204	CU-000388	BR-020	ST-48	LT-01	CT-01	15 May 2023	15 May 2028	26 August 2026	60	13.54	12817.43	290.59	11797.03	0	42.7	Yes	No	Partial Payment	Medium	Active	<30 Days
LN-001208	L-7201208	CU-000233	BR-006	ST-43	LT-01	CT-01	10 June 2020	10 June 2024	02 May 2026	48	13.86	12623.96	352.96	12125.78	0	15.54	Yes	No	Partial Payment	Medium	Active	121-150 Days
LN-001212	L-7201212	CU-000194	BR-011	ST-47	LT-01	CT-01	08 February 2023	08 February 2026	16 June 2026	36	12.53	17761.65	592.95	17285.19	0	41.3	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-001215	L-7201215	CU-000196	BR-012	ST-48	LT-01	CT-01	23 November 2023	23 November 2024	01 August 2026	12	12.43	16888.35	1512.9	15800.81	0	33.5	Yes	No	Partial Payment	Medium	Active	31-60 Days
LN-001222	L-7201222	CU-000267	BR-014	ST-42	LT-01	CT-01	26 May 2022	26 May 2023	23 June 2026	12	12.21	16323.47	1454.59	14584.83	0	48.42	Yes	No	Partial Payment	Medium	Active	91-120 Days
LN-001233	L-7201233	BR-017	ST-43	LT-01	CT-01		13 June 2021	13 June 2026	27 July 2026	60	10.41	23022.06	494.02	21782.96	0	18.58	Yes	No	Partial Payment	Medium	Active	31-60 Days

Table LoanFact (5,000 rows) (used Columns CustomerID (2,773 distinct values)

Figure 1: Loan Fact Table

This produced a classic **star-schema structure**, with LoanFact at the centre and descriptive dimensions surrounding it.

The model connects loan records to attributes such as customer segment, income band, credit score, branch, state, loan type, collateral, risk level and dates.

The project documentation identifies fields such as LoanID, CustomerID, BranchID, LoanTypeID, CollateralID, OriginalLoanAmount, TotalAmountPaid, CollateralValue, PerformanceScore, MissedPaymentFlag and DefaultFlag as key components of the analysis.

Data model

The resulting model looked like this:

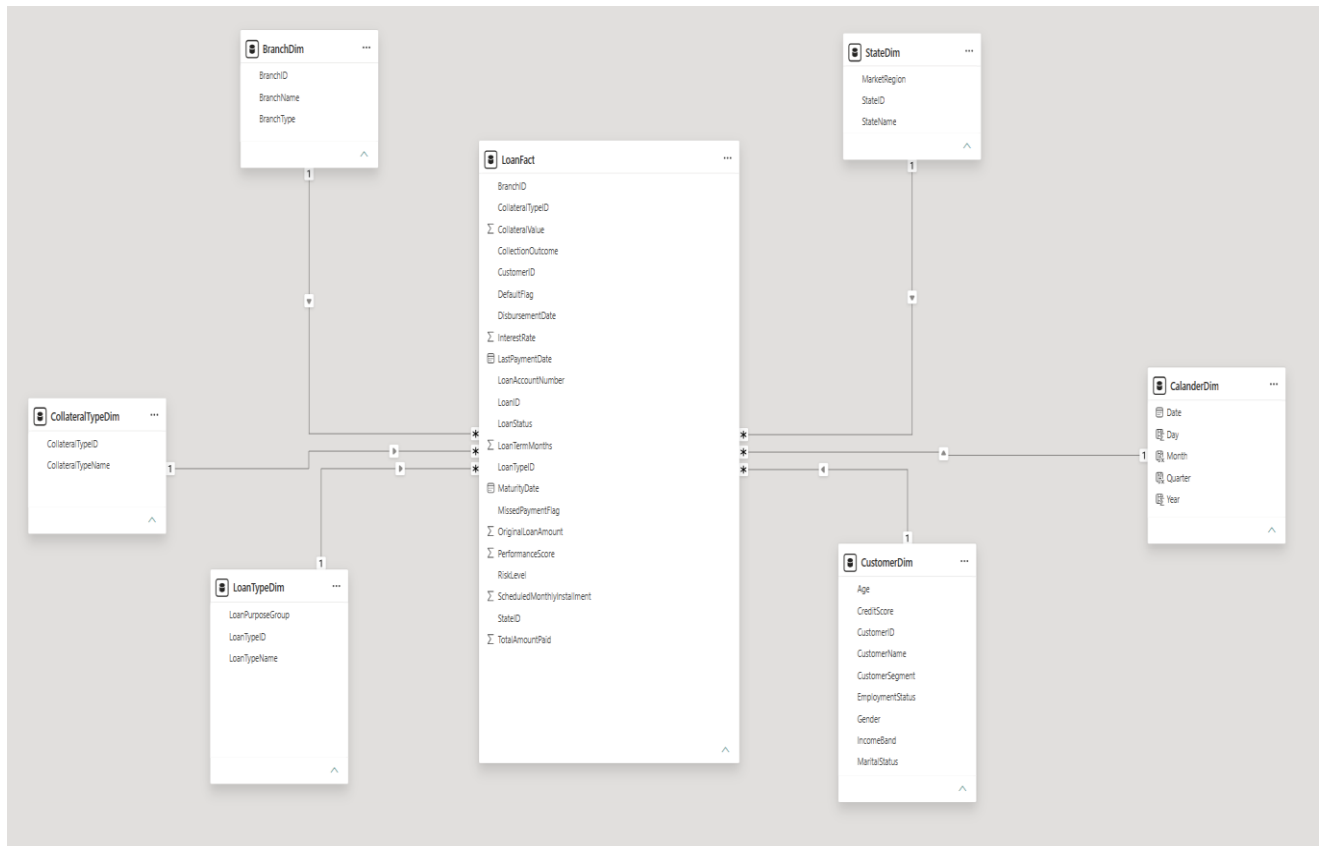


Figure 2. Star-schema data model

The project specification explicitly called for customer, loan, branch, product, geographical and collateral information to be brought into one connected structure.

Creating the Calendar Dimension

I also created a dedicated calendar table covering the analysis period.

This allowed the report to analyse lending and risk metrics across:

- Year
- Quarter
- Month
- Day

This became particularly useful for the monthly lending trend and default-rate analysis.

1 CalenderDim = CALENDAR (DATE (2020,01,01),DATE (2054,12,31))							
Date	Year	Month	Day	Quarter	Month(SF)	MonthNumber	
01/01/2020	2020	January	1	Q1	Jan	1	
02/01/2020	2020	January	2	Q1	Jan	1	
03/01/2020	2020	January	3	Q1	Jan	1	
04/01/2020	2020	January	4	Q1	Jan	1	
05/01/2020	2020	January	5	Q1	Jan	1	
06/01/2020	2020	January	6	Q1	Jan	1	
07/01/2020	2020	January	7	Q1	Jan	1	
08/01/2020	2020	January	8	Q1	Jan	1	
09/01/2020	2020	January	9	Q1	Jan	1	
10/01/2020	2020	January	10	Q1	Jan	1	
11/01/2020	2020	January	11	Q1	Jan	1	
12/01/2020	2020	January	12	Q1	Jan	1	
13/01/2020	2020	January	13	Q1	Jan	1	
14/01/2020	2020	January	14	Q1	Jan	1	
15/01/2020	2020	January	15	Q1	Jan	1	
16/01/2020	2020	January	16	Q1	Jan	1	
17/01/2020	2020	January	17	Q1	Jan	1	
18/01/2020	2020	January	18	Q1	Jan	1	
19/01/2020	2020	January	19	Q1	Jan	1	
20/01/2020	2020	January	20	Q1	Jan	1	
21/01/2020	2020	January	21	Q1	Jan	1	
22/01/2020	2020	January	22	Q1	Jan	1	
23/01/2020	2020	January	23	Q1	Jan	1	
24/01/2020	2020	January	24	Q1	Jan	1	
25/01/2020	2020	January	25	Q1	Jan	1	
26/01/2020	2020	January	26	Q1	Jan	1	
27/01/2020	2020	January	27	Q1	Jan	1	
28/01/2020	2020	January	28	Q1	Jan	1	
29/01/2020	2020	January	29	Q1	Jan	1	
30/01/2020	2020	January	30	Q1	Jan	1	
31/01/2020	2020	January	31	Q1	Jan	1	
01/02/2020	2020	February	1	Q1	Feb	2	
02/02/2020	2020	February	2	Q1	Feb	2	
03/02/2020	2020	February	3	Q1	Feb	2	
04/02/2020	2020	February	4	Q1	Feb	2	
05/02/2020	2020	February	5	Q1	Feb	2	
06/02/2020	2020	February	6	Q1	Feb	2	
07/02/2020	2020	February	7	Q1	Feb	2	
08/02/2020	2020	February	8	Q1	Feb	2	
09/02/2020	2020	February	9	Q1	Feb	2	
10/02/2020	2020	February	10	Q1	Feb	2	
11/02/2020	2020	February	11	Q1	Feb	2	
12/02/2020	2020	February	12	Q1	Feb	2	
13/02/2020	2020	February	13	Q1	Feb	2	
14/02/2020	2020	February	14	Q1	Feb	2	
15/02/2020	2020	February	15	Q1	Feb	2	
16/02/2020	2020	February	16	Q1	Feb	2	
17/02/2020	2020	February	17	Q1	Feb	2	
18/02/2020	2020	February	18	Q1	Feb	2	
19/02/2020	2020	February	19	Q1	Feb	2	
20/02/2020	2020	February	20	Q1	Feb	2	

Table: CalenderDim (12,784 rows)

Figure 3: Calender Dim Table

DAX & KPI Development

With the model in place, I created measures to turn the raw loan data into business metrics.

Some of the key measures included:

- **Total Loan Issued**

</> DAX

Total Loan Issued = SUM(LoanFact[OriginalLoanAmount])

- **Total Amount Paid**

</> DAX

Total Amount Paid = SUM(LoanFact[TotalAmountPaid])

- **Expected Interest**

The expected interest calculation was based on the original loan amount, interest rate, and loan term:

</> DAX

*Expected Interest = SUMX(LoanFact,
LoanFact[OriginalLoanAmount]
*LoanFact[InterestRate]
*LoanFact[LoanTermMonths]
/12/100)*

- **Expected Revenue**

</> DAX

Expected Revenue = [Total Loan Issued]+[Expected Interest]

- **Loan Collection Rate**

</> DAX

Loan Collection Rate = DIVIDE([Total Amount Paid],[Total Loan Issued],0)

- **Total Customers**

</> DAX

Total Customers = DISTINCTCOUNT(LoanFact[CustomerID])

- **Total Amount Unpaid**

</> DAX

Total Amount Unpaid = [Total Loan Issued]-[Total Amount Paid]

- **Default Rate**

</> **DAX**

Default Rate = DIVIDE([Defaulted Loans],COUNT(LoanFact[LoanID]), 0)

- **Customer with Defaulted Loans**

</> **DAX**

*Cust. with Defaulted Loans = CALCULATE(
DISTINCTCOUNT(LoanFact[CustomerID]),
LoanFact[DefaultFlag]="Yes")*

- **Average Days Without Payment**

</> **DAX**

*Avg. Days Without Payment = AVERAGEX(LoanFact,
DATEDIFF(LoanFact[LastPaymentDate],TODAY(),DAY))*

- **High Risk Exposure**

</> **DAX**

*High Risk Exposure = CALCULATE(
SUM(LoanFact[OriginalLoanAmount]),
LoanFact[RiskLevel]="High")*

- **Defaulted Loans**

</> **DAX**

*Defaulted Loans = CALCULATE(
COUNT(LoanFact[LoanID]),
LoanFact[DefaultFlag]="Yes")*

The project workflow specifically called for using DAX to calculate KPIs such as default rates and portfolio exposure.

DASHBOARD DESIGN

I designed two main analytical pages.

Page 1: Profit Analysis

The first page focuses on **portfolio performance and lending activity**.

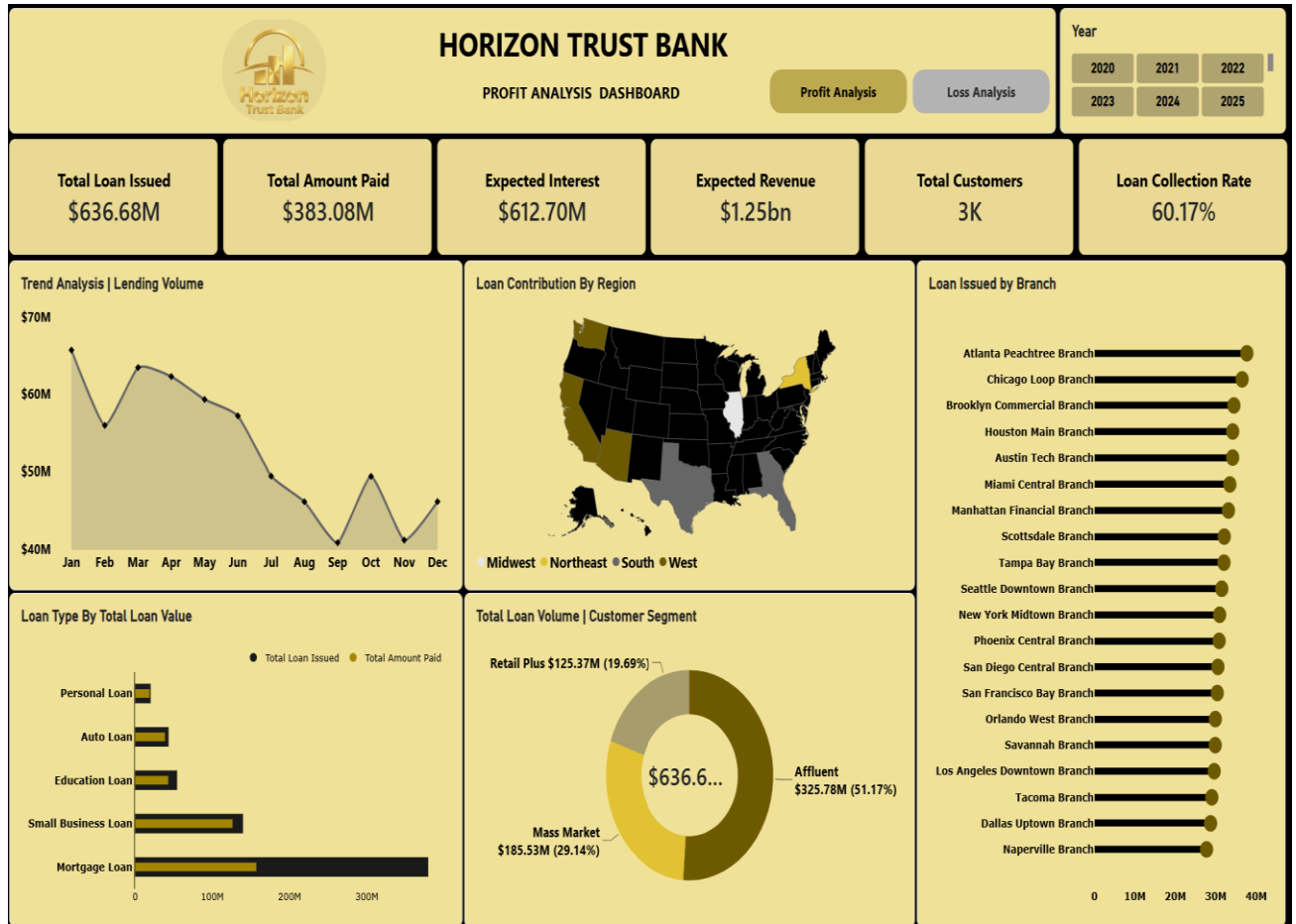


Figure 4. Profit Analysis Dashboard

Portfolio Performance (Profit)



Figure 4.1: Profit Analysis KPI

Portfolio Performance: What Did the Analysis Reveal?

1. \$636.68M Has Been Issued

The bank has issued approximately \$636.68 million across the 5,000 loan records.

This represents the scale of the lending operation and provides the starting point for understanding repayment and risk.

However, looking at the amount issued alone does not tell us how much of the portfolio has actually been recovered.

2. \$383.08M Has Been Collected

Customers have paid approximately \$383.08 million.

That represents a portfolio collection rate of:

60.17%

This means that roughly six out of every ten dollars of original loan value has been collected.

The remaining exposure is therefore an important part of the analysis.

3. Mortgage loans dominate the portfolio

Mortgage lending is by far the largest loan category. Mortgage loans account for approximately **\$379.40M** or about **59.6% of total loan value**.

Small Business Loans follow at approximately **\$139.63M**, while Education, Auto and Personal Loans contribute considerably smaller amounts.

This concentration is important because a large proportion of the bank's lending exposure is tied to a single loan category.

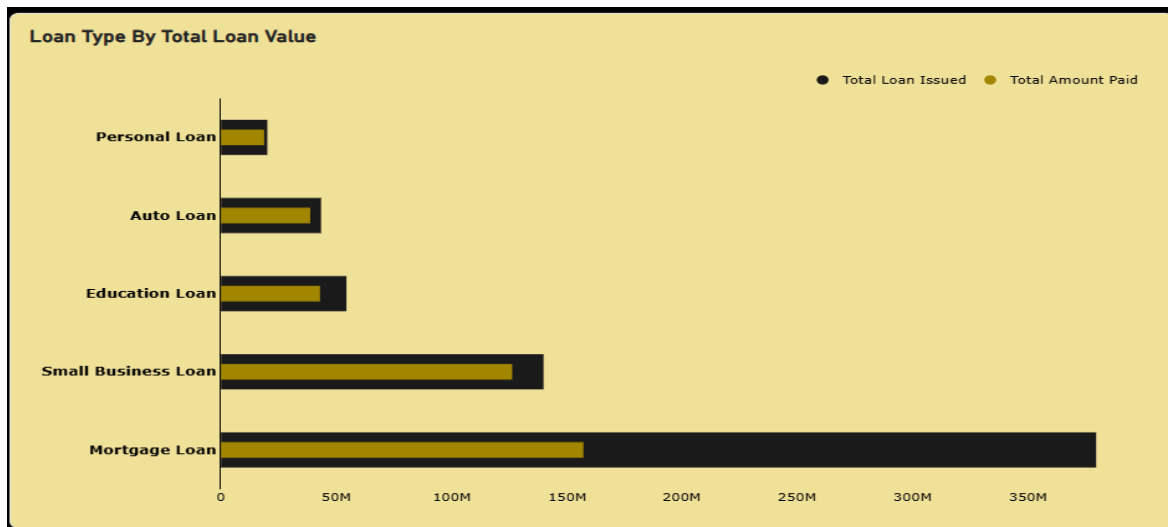


Figure 5: Total Loan Value

The issue becomes even more interesting when repayment performance is considered.

Mortgage loans have generated approximately **\$156.89M in payments**, giving them a collection rate of roughly **41.4%** considerably below the portfolio-wide collection rate of 60.17%.

That makes mortgage exposure particularly important to monitor.

4. Lending volume is not constant throughout the year

The monthly lending analysis revealed a noticeable pattern.

Loan issuance peaked in **January at approximately \$65.65M**, before falling to around **\$55.91M in February**.

The portfolio then recovered in March and April before gradually declining toward September.

There was another increase in October, followed by a November decline and December recovery. This means lending activity is not evenly distributed across the calendar.

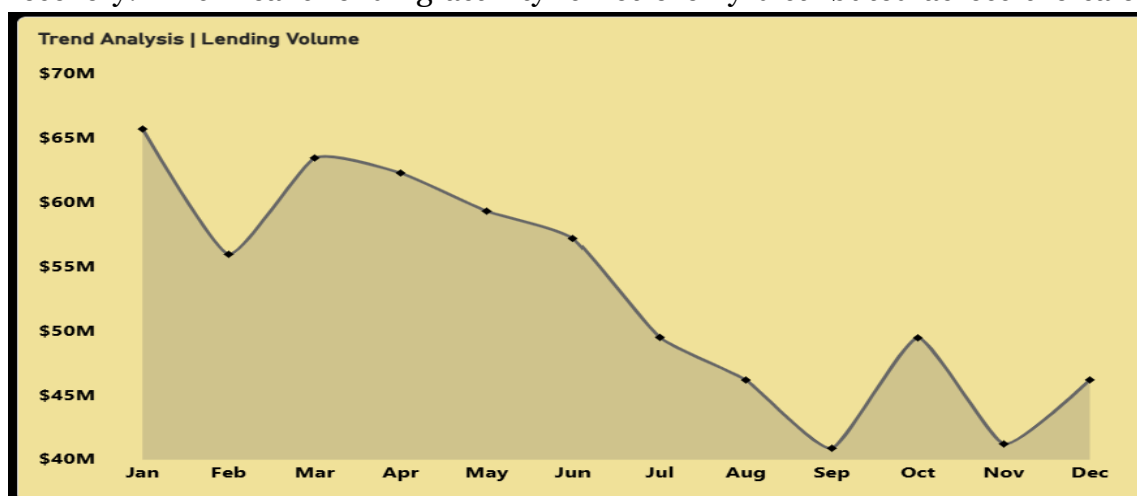


Figure 6: Lending Volume

Understanding this pattern could help the bank align staffing, loan processing capacity and collection resources with periods of higher lending activity.

5. Regional Lending Concentration

At the regional level, lending is concentrated across four market regions:

Region	Loan Value
South	\$260.00M
West	\$213.90M
Northeast	\$98.55M
Midwest	\$64.23M

The **South** therefore represents the largest regional contribution to the portfolio, followed by the West.

At state level, New York, Texas, Florida and California represent some of the largest lending markets.

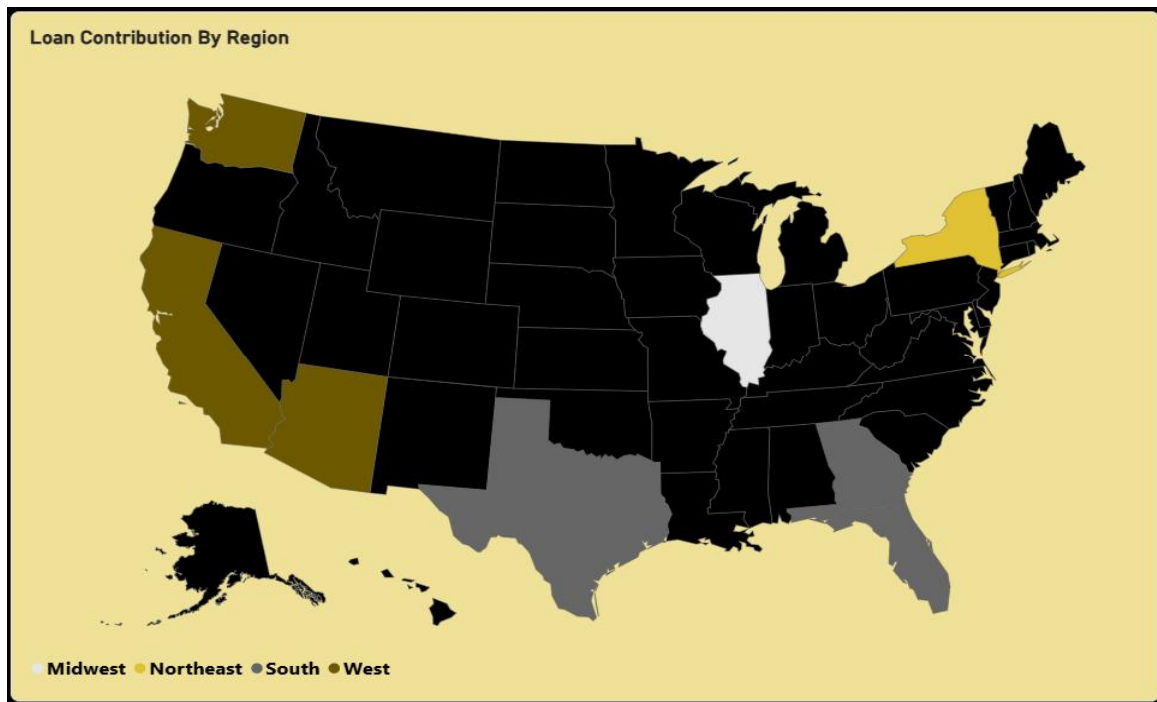


Figure 7: Loan Contribution by Region

6. Customer Segment Analysis

The portfolio is also concentrated by customer segment.

The dashboard shows:

- **Affluent - \$325.78M**
- **Mass Market - \$185.53M**
- **Retail Plus - \$125.37M**

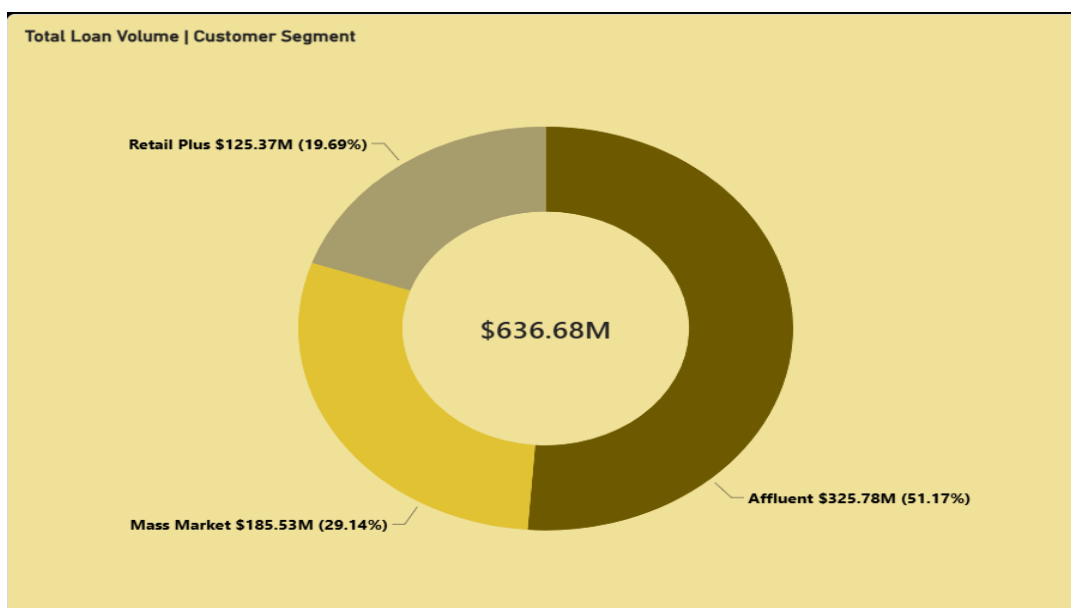


Figure 8: Customer Segment

Affluent customers therefore account for the largest loan value.

But this is where analysing only the lending side can be misleading.

A segment can represent a smaller share of loan value while still producing a larger number of defaults.

That became clear when I moved to the Loss Analysis page.

Page 2: Loss Analysis

The second dashboard page focuses on the other side of the portfolio:

What could the bank potentially lose?

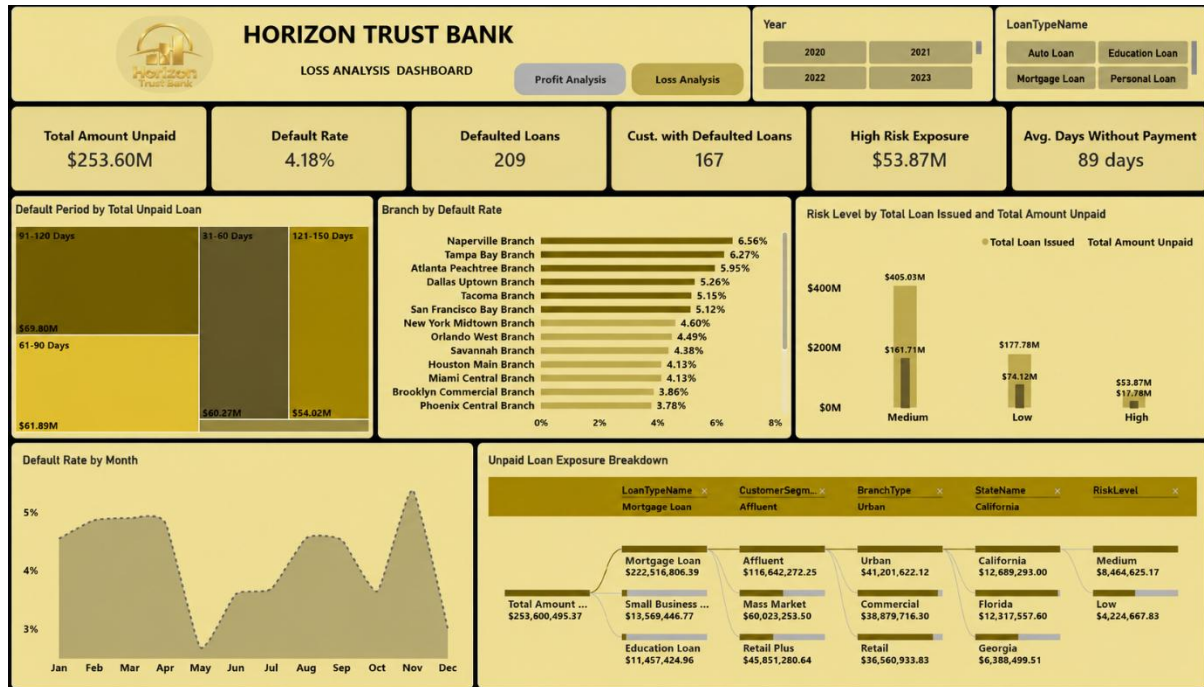


Figure 9: Loss Analysis Dashboard

Portfolio Performance (Loss)

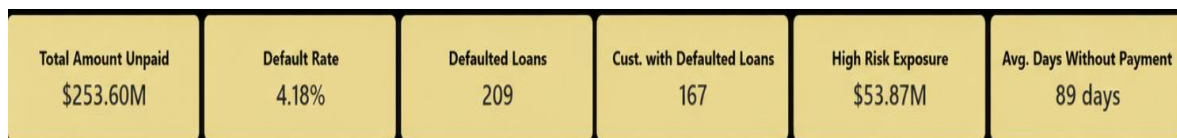


Figure 9.1: Loss Analysis KPI

The loss analysis was designed around questions including default periods, branch default rates, unpaid exposure by risk level, changes in default rates over time, and the customer, loan, risk and geographic factors contributing to default exposure.

1. \$253.60M Remains Unpaid

The difference between the amount issued and amount collected is substantial:

$$\text{\$636.68M} - \text{\$383.08M} = \text{\$253.60M}$$

So while the bank has collected more than \$383M, there is still approximately **\$253.6M in unpaid loan value**. This became the central metric for the loss analysis.

2. Default Rate

The dataset contains **209 defaulted loans** out of 5,000 loan records.

That produces a default rate of **4.18%**

There are also **167 unique customers** associated with those defaulted loans.

This distinction matters because the number of defaulted loans and the number of customers affected are not necessarily the same.

3. High-risk exposure is significant

The dashboard shows approximately **\$53.87M** in high-risk loan exposure.

When broken down by risk level, the portfolio contains:

- Low-risk loan value: approximately **\$177.78M**
- Medium-risk loan value: approximately **\$405.03M**
- High-risk loan value: approximately **\$53.87M**

Interestingly, the high-risk category contains all **209 defaulted loans** in this dataset.

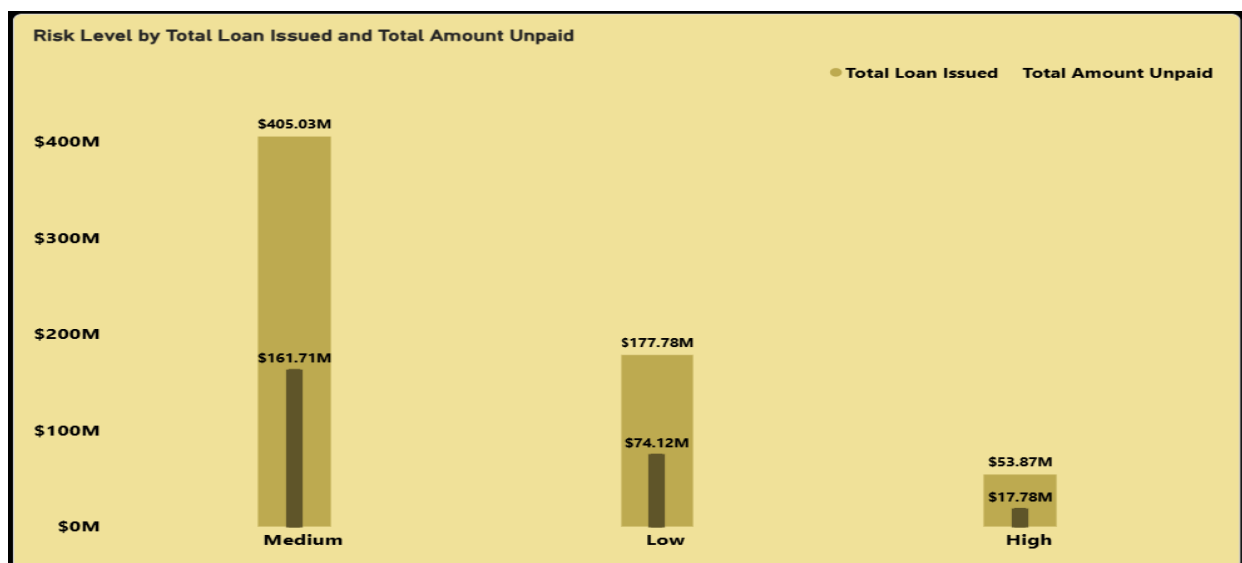


Figure 10: Risk Level

This makes the high-risk segment an important population for proactive monitoring and collection activity.

4. Unpaid exposure is concentrated in longer payment delays

The unpaid-loan analysis was particularly revealing.

Days Payment	Without Unpaid Exposure
31–60 Days	\$60.27M
61–90 Days	\$61.89M
91–120 Days	\$69.80M
121–150 Days	\$54.02M

The largest exposure sits in the **91–120 day** category, at approximately **\$69.80M**.

This shows why repayment monitoring should not only focus on whether a loan has defaulted. **How long a payment has been outstanding also matters.**

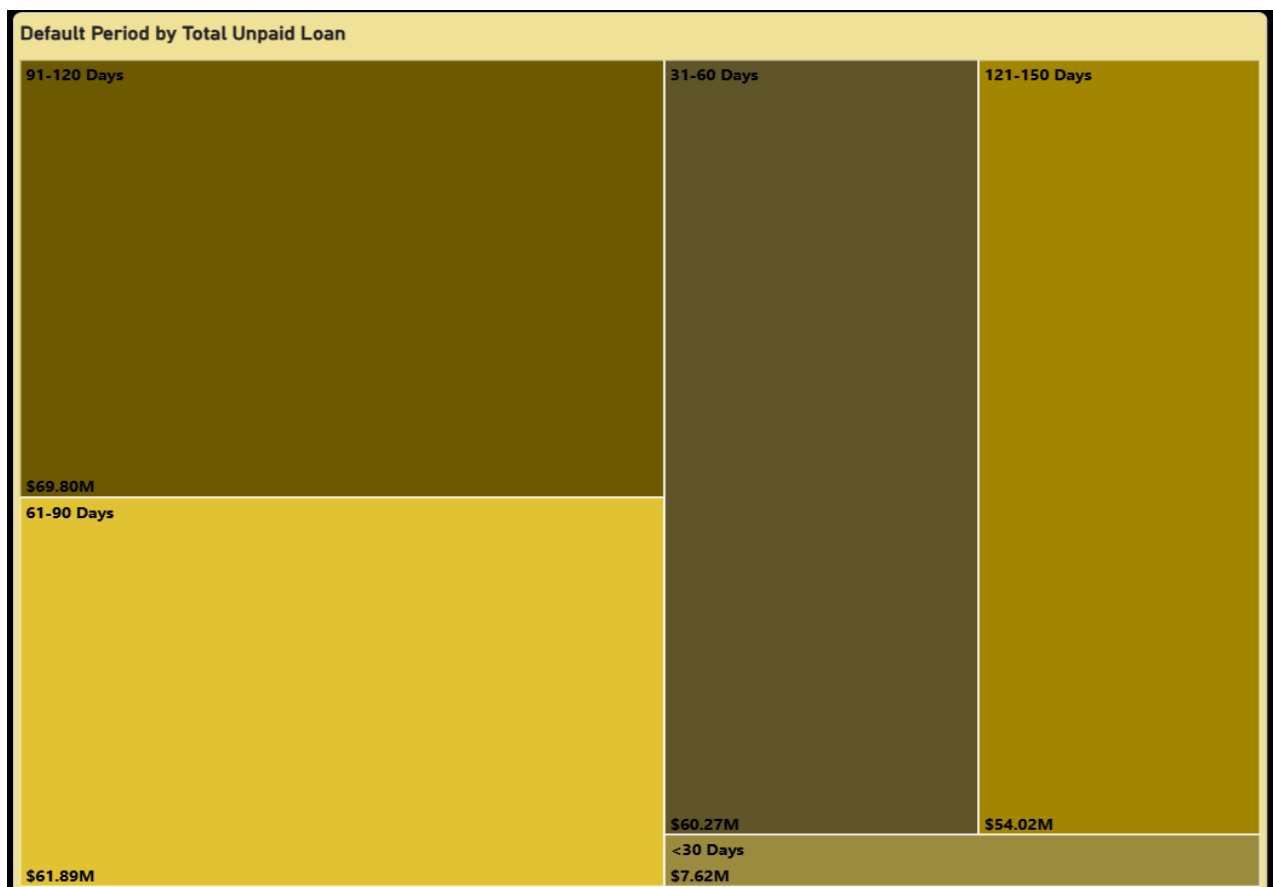


Figure 11: Default Period

5. Branch Default Rates

The branch analysis produced one of the clearest risk signals in the dashboard.

The highest default rates were recorded at:

- **Naperville Branch - 6.56%**
- **Tampa Bay Branch - 6.27%**
- **Atlanta Peachtree Branch - 5.95%**
- **Dallas Uptown Branch - 5.26%**
- **Tacoma Branch - 5.15%**
- **San Francisco Bay Branch - 5.12%**

By comparison, branches such as Austin Tech and Manhattan Financial recorded default rates below 2.5%.

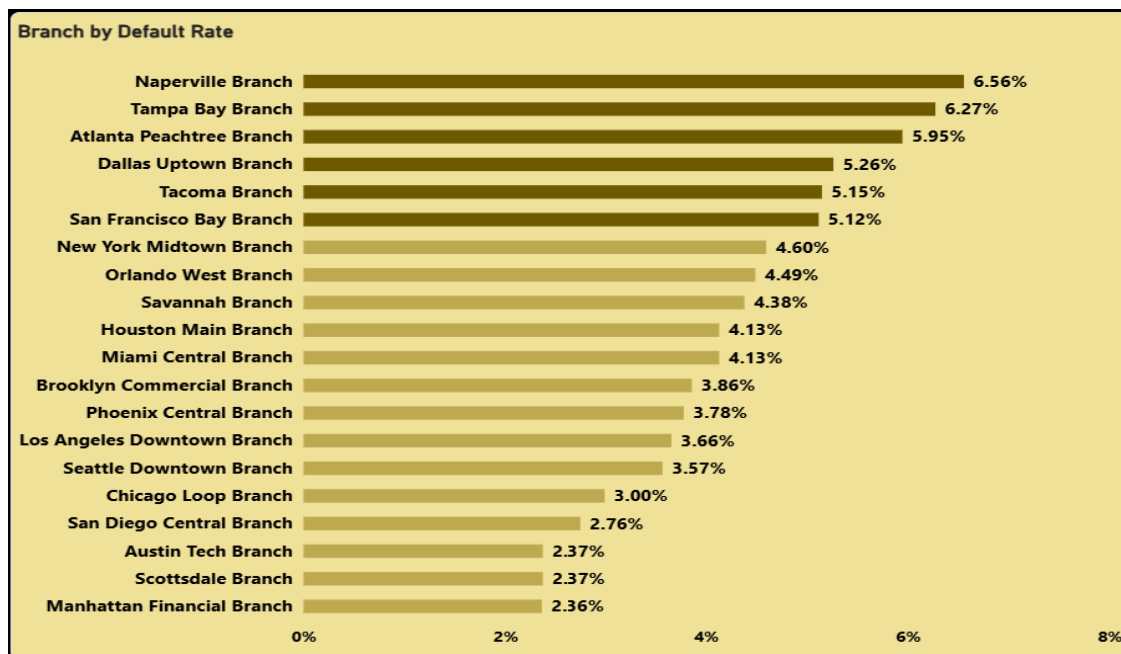


Figure 12: Branch Default Rates

These figures do not, by themselves, establish why a branch has a higher default rate.

Instead, they identify branches where a deeper investigation could be useful.

For example, management could compare these branches by:

- Loan type
- Customer segment
- Credit score
- Income band
- Employment status
- Risk level
- Collateral

- Collection outcome

This is where the interactive nature of Power BI becomes particularly valuable.

6. Mass Market customers account for most defaults

The customer-segment analysis reveals an interesting contrast.

The **Affluent** segment has the largest loan exposure at approximately **\$325.78M**.

However, the **Mass Market** segment accounts for **196 of the 209 defaulted loans**.

That's approximately **94% of all defaulted loans** in the dataset.

This doesn't mean the Mass Market segment has the largest financial exposure, the Affluent segment has a larger unpaid balance because its loans are generally larger.

Instead, it shows that **loan count and financial exposure tell different stories**.

That distinction is important when designing a credit-risk strategy.

7. The bank collected about 60% of the original loan value

Across the entire portfolio, the bank has collected **\$383.08M out of \$636.68M** giving a collection rate of approximately **60.17%**

However, collection performance varies considerably by loan type.

Loan Type	Approx. Collection Rate
Personal Loan	93.14%
Small Business Loan	90.28%
Auto Loan	89.19%
Education Loan	78.90%
Mortgage Loan	41.35%

Mortgage lending represents the majority of the bank's portfolio by value, yet it has the lowest collection rate among the five loan categories.

That combination makes mortgage loans particularly important from an exposure-management perspective.

KEY INSIGHTS

After bringing the analysis together, five major insights stood out.

1. The portfolio is highly concentrated in mortgage lending

Mortgage loans account for approximately 60% of total loan value.

2. Collection performance is materially below total lending

The bank has issued \$636.68M but collected \$383.08M, resulting in a 60.17% collection rate.

3. Unpaid exposure is substantial

Approximately \$253.60M remains unpaid.

4. Risk is concentrated in identifiable areas

Certain branches show higher default rates, while Mass Market customers account for most defaulted loans.

5. Payment delays represent an important risk signal

The largest unpaid exposure falls within the 91–120 day payment-delay period, while average days without payment is approximately 88.94 days.

RECOMMENDATIONS

Based on the analysis, several areas deserve management attention.

1. Strengthen monitoring of mortgage exposure

Mortgage loans represent nearly 60% of total loan value while showing a collection rate of approximately 41%.

Management could therefore monitor mortgage repayment performance more closely and investigate the drivers behind delayed payments.

2. Prioritize early intervention on prolonged unpaid loans

The \$69.80M sitting in the 91–120 day unpaid category demonstrates the financial significance of repayment delays.

An early-warning framework could help identify loans approaching higher-risk payment periods before exposure becomes more difficult to recover.

3. Investigate branch-level differences

Branches such as Naperville, Tampa Bay and Atlanta Peachtree have relatively high default rates.

Rather than treating branch location itself as the cause, management could investigate whether differences are associated with:

- customer segment
- loan type
- credit scores
- income bands
- employment status
- collateral
- repayment behaviour

4. Monitor Mass Market credit risk

Although Affluent customers account for the largest loan value, Mass Market customers account for the overwhelming majority of defaulted loans.

This suggests that portfolio monitoring should consider both **financial exposure** and **default frequency** rather than relying on one metric.

5. Use the dashboard for continuous portfolio monitoring

The Power BI dashboard can serve as an ongoing monitoring tool rather than a one-time report.

Management can filter by year and loan type and drill into branch, customer segment, geography and risk level to investigate changes in portfolio performance.

CONCLUSION

This project started as a loan portfolio analysis exercise, but the deeper story emerged from connecting **lending performance with repayment behaviour and credit risk**.

Horizon Trust Bank has a **\$636.68M loan portfolio**, but approximately **\$253.60M remains unpaid**. Mortgage loans account for the majority of the portfolio, while Mass Market customers account for most defaults. At the branch level, default rates vary considerably, and unpaid exposure becomes particularly significant once loans remain unpaid for several months.

The analysis also demonstrates why a banking dashboard needs to go beyond headline numbers.

A single metric such as total loans issued can tell management how much has been lent. But combining it with **collection rate, unpaid exposure, default rate, risk level, customer segment, loan type, geography and branch performance** provides a much more complete picture of portfolio health.

The final Power BI solution brings these perspectives together in two interactive pages:

Profit Analysis focused on lending volume, repayments, expected interest, expected revenue, customer segments, loan types, regions and branches.

Loss Analysis focused on unpaid exposure, defaults, risk levels, payment delays, branch default rates and the drivers of loan loss exposure.

The result is not just a dashboard.

It is a framework for asking better questions about the bank's loan portfolio.

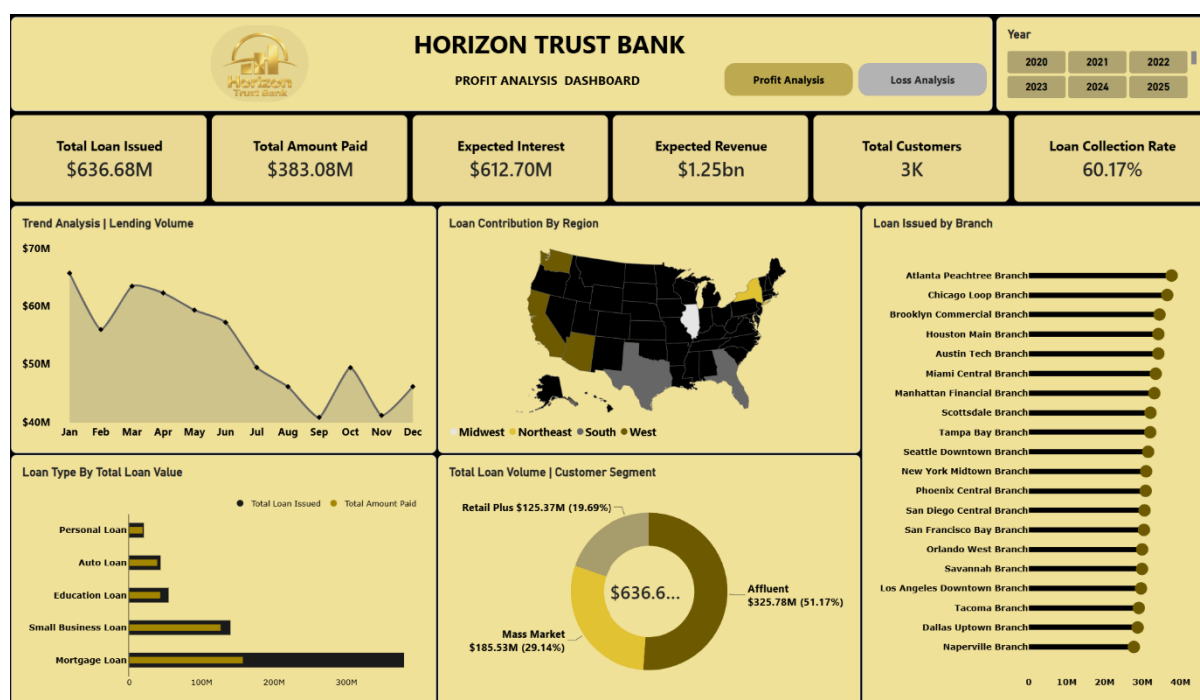
WHAT I BUILT

The final Power BI solution consists of two connected analytical report (views).

1. Profit Analysis

Provides visibility into:

- Total loan issued
- Total amount paid
- Expected interest
- Expected revenue
- Customer count
- Collection rate
- Lending trends
- Regional contribution
- Loan type performance
- Customer segment
- Branch lending

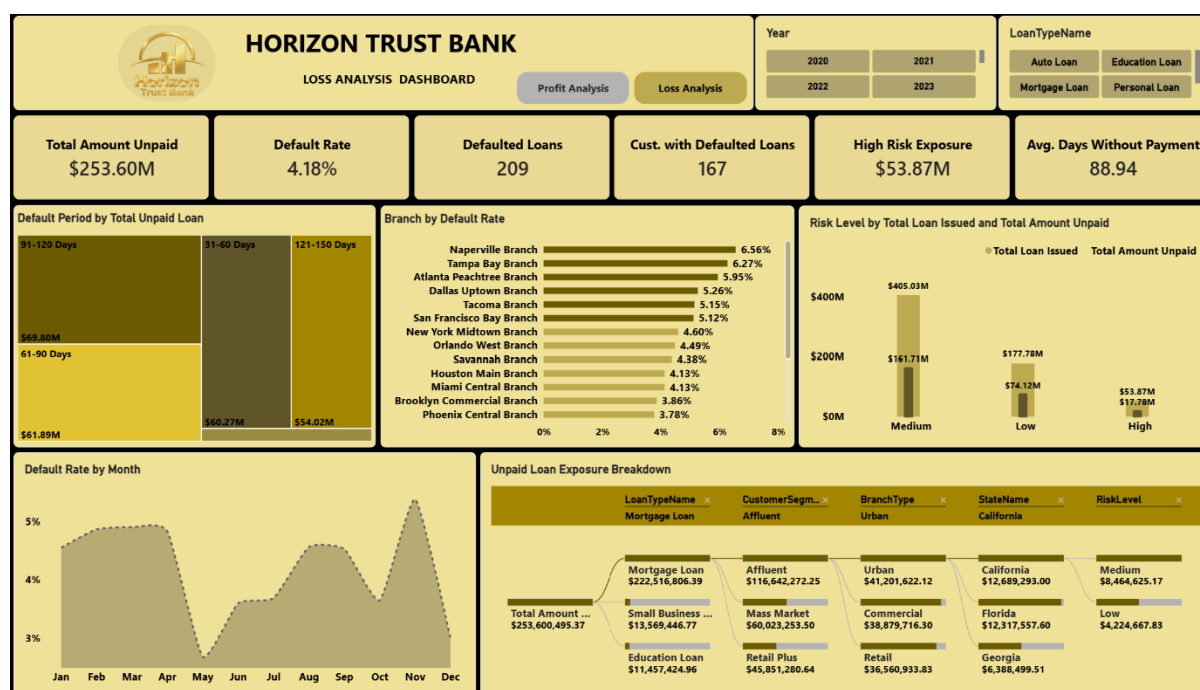


2. Loss Analysis

Provides visibility into:

- Total unpaid amount
- Default rate
- Number of defaulted loans
- Default customers
- High-risk exposure
- Average days without payment
- Default periods
- Branch default rates
- Risk-level exposure
- Unpaid exposure breakdown

Both pages can be filtered interactively by year, with the loss page also supporting loan-type filtering.



CHALLENGES & WHAT I LEARNED

This project was more than a visualization exercise.

The biggest learning for me was understanding the importance of **data modelling before dashboard design**.

A dashboard can contain attractive visuals, but if the underlying relationships are poorly structured, the analysis can quickly become unreliable or difficult to maintain.

Building the star schema gave me a much clearer understanding of how fact and dimension tables work together.

I also strengthened my ability to:

- Design analytical data models
- Create DAX measures
- Build financial KPIs
- Analyse credit-risk indicators
- Use Power BI interactions
- Translate raw data into business questions
- Present complex information through executive dashboards

The project workflow itself was structured around cleaning the data, constructing the model, creating DAX KPIs, designing the dashboard and translating the findings into actionable insights.

Tools and Skills used:

❖ Power BI

- DAX
- Data Modelling
- Star Schema
- Power Query / Data Preparation

Project Structure

Data Source → Data Cleaning → Data Model → DAX Measures → Dashboard → Analysis → Recommendations